

PORTO ALEGRE, Brazil

WMO: 839710

Lat: 29.994S

Lon: 51.171W

Elev: 11

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.5	42.8	34.8	29.6	47.1	37.4	32.9	47.7	22.8	55.3	20.3	57.3	2.8	300	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	94.8	76.2	91.6	75.2	89.4	74.6	79.4	88.9	78.2	87.0	77.0	84.9	8.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	140.9	82.9	75.5	133.8	81.3	75.0	131.1	80.8	42.9	89.4	41.6	87.2	40.5	85.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	18.4	16.2	34.6	99.5	2.5	2.4	32.9	101.2	31.4	102.6	30.0	103.9	28.3	105.7
			34.7	82.0	2.3	1.8	33.0	83.3	31.6	84.3	30.3	85.3	28.6	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.5	77.9	77.7	75.2	70.1	63.6	59.4	58.4	61.3	63.6	68.2	71.9	75.8
	DBStd	8.88	4.19	4.37	4.57	5.68	5.84	6.85	7.17	7.08	5.98	4.94	4.86	4.82
	HDD50	22	0	0	0	0	1	6	12	3	0	0	0	0
	HDD65	819	0	0	1	19	97	193	225	160	94	24	5	1
	CDD50	6788	864	776	782	602	421	289	273	354	409	563	656	799
	CDD65	2110	399	356	318	171	52	26	22	46	53	122	211	334
	CDH74	15715	3326	2841	2253	1017	273	184	204	443	377	682	1457	2658
	CDH80	5541	1279	1057	752	306	55	41	49	158	136	197	486	1025
(14) Wind	WSAvg	6.9	7.9	7.3	7.1	6.2	5.6	5.2	5.8	6.2	7.5	8.0	8.3	8.3
(15) Precipitation	PrecAvg	54.7	4.4	4.4	4.3	4.0	4.2	5.2	5.4	5.2	5.7	5.1	4.0	4.3
	PrecMax	72.4	9.4	9.9	11.9	11.6	17.3	14.4	12.2	13.0	11.7	12.1	11.3	8.8
	PrecMin	28.0	0.4	0.9	0.6	0.0	0.1	0.3	0.7	0.5	0.6	0.8	0.3	1.2
	PrecStd	9.3	2.1	1.9	2.2	2.5	3.0	3.0	2.7	2.8	2.6	3.0	2.4	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.9	97.1	94.9	91.6	85.9	84.5	85.6	89.9	91.8	91.5	93.6	98.3
		MCWB	77.2	78.7	76.5	75.1	72.2	69.9	67.1	68.2	70.6	73.5	73.4	76.0
	2%	DB	93.1	93.0	90.6	87.6	80.6	80.2	80.2	84.6	82.8	84.7	89.4	93.1
		MCWB	76.2	77.7	75.8	74.0	68.7	68.5	66.1	67.7	69.2	71.8	72.7	75.1
	5%	DB	89.7	89.4	87.5	83.7	76.8	73.8	74.4	78.8	77.3	80.8	85.7	89.4
		MCWB	75.6	76.3	74.6	72.2	67.7	65.9	64.8	66.8	67.2	69.7	70.9	74.0
	10%	DB	86.7	86.1	84.2	79.7	73.2	69.6	69.4	73.2	73.3	77.2	82.1	85.8
		MCWB	74.9	75.3	73.5	70.5	66.3	64.3	63.1	64.3	65.3	68.2	69.9	73.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.4	81.7	80.0	77.7	73.5	71.5	70.0	71.6	73.2	76.2	77.3	79.8
		MCDB	90.6	91.9	88.4	86.9	82.0	81.0	79.7	82.2	84.6	85.2	86.4	90.4
	2%	WB	78.7	79.6	78.1	75.7	70.8	69.2	67.6	69.0	70.3	73.6	74.9	77.9
		MCDB	89.2	88.8	85.9	83.2	76.4	76.5	75.4	78.9	79.2	82.0	83.8	87.3
	5%	WB	77.4	78.2	76.4	73.9	69.1	67.2	65.7	67.2	68.6	71.5	73.2	76.3
		MCDB	86.2	86.4	83.4	80.2	74.0	72.8	72.4	76.6	76.1	78.2	80.5	84.8
	10%	WB	76.2	76.9	75.0	72.3	67.4	65.2	64.1	65.4	66.9	69.9	71.8	74.9
		MCDB	83.8	83.9	81.3	77.4	71.7	69.0	69.3	72.5	72.2	75.0	78.7	82.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	16.5	16.9	17.1	16.2	16.7	17.2	18.6	16.6	16.1	18.2	18.3
		MCDBR	21.9	20.8	21.6	22.6	21.8	23.9	25.5	28.7	25.3	23.0	24.7	24.5
		MCWBR	8.1	7.9	8.4	9.8	10.8	12.7	12.8	13.6	12.3	10.8	10.2	9.4
		MCDBR	18.4	18.0	17.6	17.5	16.9	20.4	20.5	24.4	20.8	18.2	19.1	20.0
	5% WB	MCWBR	7.7	7.9	8.0	8.5	9.5	11.3	11.7	12.4	10.8	9.5	9.3	8.8
(40) Clear-Sky Solar Irradiance	taub		0.408	0.411	0.400	0.381	0.367	0.382	0.388	0.476	0.522	0.453	0.414	0.412
	taud		2.416	2.409	2.431	2.447	2.459	2.416	2.363	2.060	1.949	2.206	2.340	2.373
	Ebn at Noon		296	289	282	271	259	245	248	235	241	274	292	296
	Edh at Noon		40	39	36	32	28	28	31	46	57	47	43	42
(44) All-Sky Solar Radiation	RadAvg		2030	1858	1570	1239	915	771	847	1050	1265	1574	1966	2070
	RadStd		141	121	107	100	86	57	97	90	124	173	206	166

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page